Week 12
الأسبوع 11: من - إلى

Subject/:الموضوع Chemistry **Unit Title : عنوان الوحدةReactivity 1: Energetics** **Grade 11 DP (Year 1)**

Lessonsالحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiationالتمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1: الحصة 1 **Standard enthalpy change and calorimetry (R1.1.4)** التغير في المحتوى الحراري القياسي وقياس الحرارة	• Students will apply Q = mcΔT and ΔH = −Q/n to calculate the enthalpy change of a reaction from calorimetry data.	2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 3.3 Guide processing with explicit steps, chunking, and interactive models. 6.4 Support progress monitoring with rubrics, checklists, and visual progress tools.	• Video: Calorimetry – Measuring Enthalpy Change (Save My Exams) • Notes: R1.1.4 Standard enthalpy change (ChemistryStudent) • Online quiz: Measuring enthalpy changes (iPassed) • **Practice Sheet 1 – Calorimetry (Teams)**
Lesson 2 **الحصة الثانية** **Bond enthalpies (R1.2.1)** طاقات الروابط	• Students will apply average bond enthalpy data to calculate the enthalpy change of a reaction.	2.2 Clarify syntax/structure and highlight relationships in text, formulas, diagrams. 3.2 Highlight big ideas, patterns, and critical features using organizers and examples. 8.2 Vary challenge level with differentiated tools, resources, and difficulty.	• Videos: R1.2.1 Bond enthalpy (RevisionDojo) • Notes: R1.2.1 Bond enthalpy (ChemistryStudent) • Data booklet values and notes (ibchem.com) • **Practice Sheet 2 – Bond enthalpies (Teams)**
Lesson 3 **الحصة الثالثة** **Hess's law (R1.2.2)** قانون هس	• Students will apply Hess's law to calculate enthalpy changes in multistep reactions.	3.3 Guide processing with explicit steps, chunking, and interactive models. 3.4 Support transfer using checklists, organizers, and recurring practice. 8.2 Vary challenge level with differentiated tools, resources, and difficulty.	• Videos: R1.2.2 Hess's law (RevisionDojo) • Videos: Hess' law introduction and cycles (PMT) • Video: Hess's law and bond enthalpy calculations (Save My Exams) • Notes: R1.2.2 Hess's law (ChemistryStudent) • **Practice Sheet 3 – Hess's law (Teams)**
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	Practice Sheets 1–3 and the answer key are for the class Microsoft Teams channel; students can also use the video links above on their own devices. Remind students to bring a calculator and the IB data booklet (c of water and average bond enthalpies are given in it). Eye protection and care with flammable fuels apply if the combustion calorimetry practical is run. [Add homework or assessment due date]		

Week # 11 الأسبوع#

Technology Integration دمج التكنولوجيا	Class Microsoft Teams channel for lessons, video links and practice sheets; students access them on their own devices. Video lessons (Save My Exams, RevisionDojo) and online quiz practice (iPassed).
Cross-curricular links الروابط بين المواد الدراسية	Physics: heat transfer, specific heat capacity and conservation of energy. Biology: energy content of food and respiration. Mathematics: rearranging equations, units and significant figures.
External Assessment Integration دمج التقييمات الخارجية	IB exam-style calculation questions on calorimetry, average bond enthalpies and Hess's law, using the data booklet. Links to the Internal Assessment: Tool 1, Inquiry 1–3 (investigating enthalpy changes experimentally).
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	ATL: Thinking skills (applying and analysing data), Self-management (organising multistep calculations with a checklist), Communication (showing working, units and signs clearly). Wellbeing: break long calculations into steps and ask for help early.
Standards المعايير	• R1.1.4: The standard enthalpy change for a chemical reaction, $\Delta H°$, refers to the heat transferred at constant pressure under standard conditions and states. It can be determined from the change in temperature of a pure substance. Apply the equations $Q = mc\Delta T$ and $\Delta H = -Q/n$ in the calculation of the enthalpy change of a reaction. • R1.2.1: Bond-breaking absorbs and bond-forming releases energy. Calculate the enthalpy change of a reaction from given average bond enthalpy data. • R1.2.2: Hess's law states that the enthalpy change for a reaction is independent of the pathway between the initial and final states. Apply Hess's law to calculate enthalpy changes in multistep reactions.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	Tool 1, Inquiry 1, 2, 3: How can the enthalpy change for combustion reactions, such as for alcohols or food, be investigated experimentally? Why do calorimetry experiments typically measure a smaller change in temperature than is expected from theoretical values? Structure 2.2: How would you expect bond enthalpy data to relate to bond length and polarity? TOK: To what extent can an average value, such as an average bond enthalpy, be considered knowledge about a particular molecule?
Success Criteria معايير النجاح **(Measurable)**	**Must (DOK 1–2):** I can state that ΔH is measured in kJ mol^{-1} and use $Q = mc\Delta T$ and $\Delta H = -Q/n$ to calculate an enthalpy change from calorimetry data. **Should (DOK 2–3):** I can calculate an enthalpy change from average bond enthalpy data and apply Hess's law to multistep reactions, with correct signs and units. **Could (DOK 3–4):** I can evaluate why calorimetry experiments show a smaller temperature change than theory predicts and why average bond enthalpies differ from experimentally measured values.
National Identity Domain مجالات الربط بالهوية الوطنية	Belonging الإنتماء — Global Understanding التفاهم العالمي — Choose an item. المجال Area: Innovation and Infrastructure التطبيق Implementation: Students use enthalpy calculations to compare the energy released by fuels and food, linking to innovation in energy use.